

Atividade 24

Print da atividade 12.6: Extração e leitura de metadados em arquivos digitais utilizando a ferramenta exiftool

```
(root@cleydson)-[/home/cleydson/Documentos/Arquivos]
# ls
certificate.crt private.key request.csr

(root@cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool IPTCparal.jpg
Error: File not found - IPTCparal.jpg

(root@cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool certificate.crt
ExifTool Version Number      : 13.44
File Name                    : certificate.crt
Directory                    : .
File Size                    : 1306 bytes
File Modification Date/Time   : 2026:03:25 13:56:36-03:00
File Access Date/Time        : 2026:03:25 16:23:50-03:00
File Inode Change Date/Time   : 2026:03:25 13:56:36-03:00
File Permissions              : -rw-r--r--
File Type                    : TXT
File Type Extension           : txt
MIME Type                    : text/plain
MIME Encoding                 : us-ascii
Newlines                     : Unix LF
Line Count                   : 22
Word Count                   : 24

(root@cleydson)-[/home/cleydson/Documentos/Arquivos]
# wget https://www.google.com/images/branding/googlelogo/1x/googlelogo_color_272x92dp.png -O teste.png
exiftool teste.png
--2026-03-26 15:09:37-- https://www.google.com/images/branding/googlelogo/1x/googlelogo_color_272x92dp.png
Resolvendo www.google.com (www.google.com)... 2001:4860:4829:7700::, 2001:4860:4826:7700::, 2001:4860:482c:7700::, ...
Conectando-se a www.google.com (www.google.com)[2001:4860:4829:7700::]:443... conectado.
A requisição HTTP foi enviada, aguardando resposta... 200 OK
Tamanho: 5969 (5,8K) [image/png]
Salvando em: "teste.png"

teste.png 100%[=====] 5,83K --.-KB/s em 0s

2026-03-26 15:09:38 (41,4 MB/s) - "teste.png" salvo [5969/5969]

ExifTool Version Number      : 13.44
File Name                    : teste.png
Directory                    : .
File Size                    : 6.0 kB
File Modification Date/Time   : 2019:10:22 15:30:00-03:00
File Access Date/Time        : 2026:03:26 15:09:38-03:00
File Inode Change Date/Time   : 2026:03:26 15:09:38-03:00
File Permissions              : -rw-r--r--
File Type                    : PNG
File Type Extension           : png
MIME Type                    : image/png
Image Width                  : 272
Image Height                  : 92
Bit Depth                    : 8
Color Type                    : RGB with Alpha
Compression                  : Deflate/Inflate
Filter                       : Adaptive
Interlace                    : Noninterlaced
Image Size                   : 272x92
Megapixels                   : 0.025

(root@cleydson)-[/home/cleydson/Documentos/Arquivos]
#
```

Print da atividade 12.7: Execução de limpeza total de metadados para proteção de privacidade com a ferramenta exiftool

```

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool -all= teste.png
0 image files updated
1 image files unchanged

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool teste.png
ExifTool Version Number      : 13.44
File Name                    : teste.png
Directory                   : .
File Size                   : 6.0 kB
File Modification Date/Time  : 2019:10:22 15:30:00-03:00
File Access Date/Time       : 2026:03:26 15:09:38-03:00
File Inode Change Date/Time  : 2026:03:26 15:09:38-03:00
File Permissions             : -rw-r--r--
File Type                   : PNG
File Type Extension         : png
MIME Type                   : image/png
Image Width                 : 272
Image Height                : 92
Bit Depth                   : 8
Color Type                  : RGB with Alpha
Compression                 : Deflate/Inflate
Filter                      : Adaptive
Interlace                   : Noninterlaced
Image Size                  : 272x92
Megapixels                  : 0.025

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# wget https://raw.githubusercontent.com/ianare/exif-samples/master/jpg/gps/DSCN0010.jpg -O foto.jpg
--2026-03-26 15:14:57-- https://raw.githubusercontent.com/ianare/exif-samples/master/jpg/gps/DSCN0010.jpg
Resolvendo raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8003::154, 2606:50c0:8000::154, 2606:50c0:8001::154, ...
Conectando-se a raw.githubusercontent.com (raw.githubusercontent.com)|2606:50c0:8003::154|:443 ... conectado.
A requisição HTTP foi enviada, aguardando resposta... 200 OK
Tamanho: 161713 (158K) [image/jpeg]
Salvando em: "foto.jpg"

foto.jpg 100%[=====] 157,92K --.-KB/s em 0,07s

2026-03-26 15:14:57 (2,13 MB/s) - "foto.jpg" salvo [161713/161713]

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool -all= foto.jpg
1 image files updated

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool foto.jpg
ExifTool Version Number      : 13.44
File Name                    : foto.jpg
Directory                   : .
File Size                   : 146 kB
File Modification Date/Time  : 2026:03:26 15:15:04-03:00
File Access Date/Time       : 2026:03:26 15:15:04-03:00
File Inode Change Date/Time  : 2026:03:26 15:15:04-03:00
File Permissions             : -rw-r--r--
File Type                   : JPEG
File Type Extension         : jpg
MIME Type                   : image/jpeg
Image Width                 : 640
Image Height                : 480
Encoding Process             : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components             : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:2 (2 1)
Image Size                  : 640x480
Megapixels                  : 0.307

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
#

```

Print da atividade 12.8: Utilização da ferramenta exiftool para a edição e inserção de novos metadados em arquivos digitais

```
root@Cleydson: /home/cleydson/Documentos/Arquivos

Sessão  Ações  Editar  Exibir  Ajuda

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool -rights="Direitos autorais de Cleydson" -CopyrightNotice="Direitos autorais de Cleydson" foto.jpg
1 image files updated

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool -Artist="Cleydson" foto.jpg
1 image files updated

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool "-DateTimeOriginal=2017:12:12 06:00:00" foto.jpg
1 image files updated

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
# exiftool foto.jpg
ExifTool Version Number      : 13.44
File Name                    : foto.jpg
Directory                    : .
File Size                    : 150 kB
File Modification Date/Time   : 2026:03:26 15:18:02-03:00
File Access Date/Time        : 2026:03:26 15:18:02-03:00
File Inode Change Date/Time   : 2026:03:26 15:18:02-03:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Artist                       : Cleydson
Y Cb Cr Positioning          : Centered
Exif Version                  : 0232
Date/Time Original           : 2017:12:12 06:00:00
Components Configuration     : Y, Cb, Cr, -
Color Space                   : Uncalibrated
Current IPTC Digest          : bd57cc7aaba7fb26edec9a404ebfc818
Copyright Notice              : Direitos autorais de Cleydson
Application Record Version    : 4
XMP Toolkit                   : Image::ExifTool 13.44
Rights                       : Direitos autorais de Cleydson
Image Width                  : 640
Image Height                 : 480
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:2 (2 1)
Image Size                   : 640x480
Megapixels                   : 0.307

(root@Cleydson)-[/home/cleydson/Documentos/Arquivos]
```

Print da atividade 12.9: Utilização da ferramenta The Sleuth Kit (TSK) no Kali Linux para análise forense e extração de metadados de inodes

```
++ r/r 13767159: VPS.hsmiths.com
++ r/r 13767137: eai.coincited.net
++ d/d 13767143: cache
++ s/h 13767090: daemon_rpc_socket
++ d/d 13767100: wallets
++ r/r 13767166: default_wallet
++ r/r 13767151: recent_servers
d/d 8519681: root
+ d/d 8519682: Documentos
++ d/d 8519683: hdb_lab
++ r/r 8519684: Texto.txt
++ r/r 8519685: Texto_copia.txt
++ d/d 8519686: PastaB
+ r/r 8519687: cofre.img
d/d 10747905: teste1
+ r/r 10747906: .bashrc.original
+ d/d 10747907: .java
++ d/d 10747908: .userPrefs
++ d/d 10747909: burp
++++ r/r 10747910: prefs.xml
+ r/r 10747911: .zprofile
+ r/r 10747912: .face
+ d/d 10747913: .local
++ d/d 10747914: share
++ d/d 10747915: nautilus
++++ d/d 10747916: scripts
++++ r/r 10747917: Terminal
++ d/d 10747918: bin
+ r/r 10747919: .bash_logout
+ l/l 10747920: .face.icon
+ r/r 10747921: .zshrc
+ r/r 10747922: .profile
+ d/d 10747923: .config
++ d/d 10747924: nautilus
++ r/r 10747925: scripts-accel
++ d/d 10747926: xfce4
++ d/d 10747927: panel
++++ r/r 10747928: genmon-15.rc
++ d/d 10747929: autostart
++ r/r 10747930: fix-old-genmon-config.desktop
++ d/d 10747931: powershell
++ r/r 10747932: Microsoft.PowerShell_profile.ps1
++ d/d 10747933: cherrytree
++ r/r 10747934: config.cfg
+ r/r 10747935: .bashrc
V/V 24739841: $OrphanFiles

(root@Cleydson) [/home/cleydson/Documentos/Arquivos]
# ls -l /dev/nvme1n1p4 2
class|host|device|start_time
ls|Cleydson||1774549341
st_ino|st_alloc|st_uid|st_gid|st_mtime|st_atime|st_ctime|st_crtime|st_mode|st_nlink|st_size

(root@Cleydson) [/home/cleydson/Documentos/Arquivos]
#
```

Print da atividade 12.10: Realização de aquisição de imagem de disco bit-a-bit utilizando a ferramenta dd para fins de preservação forense

```
root@Cleydson: /home/cleydson/Documentos/Arquivos

Sessão Ações Editar Exibir Ajuda
root@Cleydson: /home/cleydson/Documentos/Arquivos
# cd /home/cleydson/Documentos/Arquivos
root@Cleydson: /home/cleydson/Documentos/Arquivos
# dd if=/dev/nvme1n1p4 of=Backup_nvme1n1p4 bs=1M count=100
100+0 registros entrados
100+0 registros saídos
104857600 bytes (105 MB, 100 MiB) copiados, 0,0573554 s, 1,8 GB/s

root@Cleydson: /home/cleydson/Documentos/Arquivos
# stat Backup_nvme1n1p4
Arquivo: Backup_nvme1n1p4
Tamanho: 104857600 Blocos: 204800 bloco de E/S: 4096 regular file
Dispositivo: 259,4 Inode: 13772328 Ligações: 1
Acesso: (0644/-rw-r--r--) Uid: ( 0/ root) Gid: ( 0/ root)
Acesso: 2020-03-20 15:23:51.290396078 -0300
Modificação: 2020-03-20 15:23:51.246396766 -0300
Alteração: 2020-03-20 15:23:51.246396766 -0300
Criação: 2020-03-20 15:23:51.290396078 -0300

root@Cleydson: /home/cleydson/Documentos/Arquivos
#
```